

Hollow-Core Fibre Transmission Systems: From Metro to Long-Haul Applications

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Abstract We review recent advances in high-capacity optical transmission that exploit the unique beneficial properties of hollow-core fibres (HCFs). Single-span metro, multi-span long-distance, and long-haul transmission scenarios are discussed, together with opportunities and challenges towards future HCF-based transmission systems. ©2026 The Author(s)

Introduction

The sustained growth in cloud services and artificial intelligence cluster training has been intensifying significant demands for optical transmission systems to deliver high-capacity data communication at low cost and low latency [1, 2]. However, the conventional solid-core single-mode fibres (SMFs) exhibit inherent limitations of fibre propagation latency and fibre nonlinearity, which become increasingly relevant as optical systems evolve towards higher baud rates, wider bandwidths, and longer amplifier spacings [1, 2].

In contrast, hollow-core fibre (HCF), and particularly the double nested anti-resonant nodeless fibre (DNANF) design, is emerging as a practical transmission medium for next-generation optical communications. As light is predominantly guided in the air core, HCFs offer ultimate-low latency, orders-of-magnitude lower fibre nonlinearity, low and flat chromatic dispersion (CD), and low Rayleigh backscattering [3, 4]. Moreover, recent progresses in anti-resonant HCF designs have produced ultra-low fibre attenuations over ultra-wide bandwidths, which are well below the theoretical minimum of silica-core SMFs. For example, fibre losses <0.1 dB/km were demonstrated over 1481 nm to 1625 nm (18 THz) in a single HCF [3], and fibre losses as low as 0.05 dB/km have been reported in the C-band [4].

These advances, in turn, have stimulated significant interests in system-level evaluations of HCF transmission. To date, experimental and field-oriented studies have examined the performance of HCF-enabled systems across a broad set of scenarios, encompassing short-reach (few km) interconnects [5, 6], metro-scale point-to-point links extending to approximately 120 km [1, 7-11], regional and long-distance routes reaching several hundreds of kilometres [12-17], and ultimately, long-haul transmission configurations spanning thousands of kilometres [18-24].

In this paper, we review recent progresses in high-capacity optical transmission using HCFs. We discuss single-span metro transmission, multi-span long-distance systems and long-haul HCF links, and highlight our recent work in these

scenarios. In addition, we also discuss the challenges and opportunities for leveraging the disruptive HCF technology to reshape next-generation metro, terrestrial, and long-haul networks.

Single-Span Metro Transmission

To date, most of the HCF-based transmission works adopted the single-span system configuration, which can be broadly divided into short-reach studies and metro demonstrations. For short-reach HCF investigations, one of the key advantages that HCFs offer is their low CD. This is especially beneficial for intensity modulation and direct detection (IM/DD) systems, as the CD-induced power fading represents the dominant limiting factor in such systems. The low CD offered by HCFs can be translated into higher baud rate and/or longer transmission distance for IM/DD transmission [5, 6]. For example, dense wavelength-division multiplexing (DWDM) transmission of 112-GBaud 4-ary pulse amplitude modulation signals over super C-band has been achieved in a 10-km HCF link [6], which constitutes the highest capacity per fibre core amongst recent demonstrations of short-reach IM/DD optical interconnects.

On the other hand, extending the transmission reach to the hundred-kilometre scale represents a critical building block for metro architectures and data centre interconnections. Fig. 1(a) shows the typical configuration of metro WDM transmission systems. Such systems are normally unrepeatable, with only a booster amplifier (BA) and a pre-amplifier (PA) to support single-span transmission. The transmitters (Tx) operating at different wavelengths are multiplexed through a multiplexer (MUX), and the combined WDM optical signals are boosted by the BA. At the receiver (Rx), the de-multiplexer (De-MUX) can be used to separate different wavelengths for signal reception at the corresponding Rx.

Typically, the span lengths of such metro WDM transmission links are up to around 80 to 120 km. This reach limit is mainly set by the required receiver optical signal-to-noise ratio

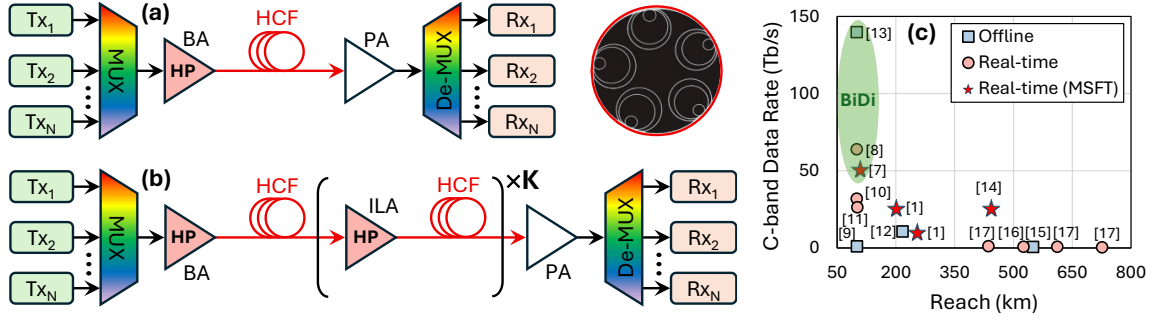


Fig. 1: (a) Typical setup of WDM single-span metro transmission, (b) setup of WDM multi-span long-distance transmission, (c) comparison of recent demonstrations of ≥ 100 -km straight-line HCF transmission.

(OSNR), which is determined by the launch power and overall link loss. When SMF is adopted, the launch power must be restricted to a medium level to avoid exciting fibre nonlinearity which will inevitably degrade the transmission performance. In contrast, HCFs can allow significantly higher launch powers without compromising the transmission performance, thanks to its ultra-low nonlinearity. This allows for the use of high-power (HP) amplifiers to either boost the receiver OSNR and/or extending the span lengths of unrepeat transmission. In the literature, optical transmission over HCFs with launch powers up to ~ 40 dBm have been reported [1, 2, 14-19]. While most of the prior HP HCF transmissions were demonstrated over either single wavelength [15-17] or relatively narrow bandwidths [18], we have achieved full C-band real-time transmission over > 100 -km of HCF with up to 37-dBm launch power [1]. Specifically, a decoded data rate of 25.6 Tb/s (32×800 Gb/s/ λ) was achieved for 102.3 km to 200.5 km using 34.5-dBm launch power to compensate for the 46-dB link loss at 200.5 km. When increasing the launch power to 37 dBm, a total data rate of 10.1 Tb/s (32×315.8 Gb/s/ λ) was also achieved over 250 km with link loss of 57 dB.

It is also worth noting that beyond real-time transmission demonstrations, we also showcased the HCF-compatible optical time-domain reflectometer, which was designed to compensate for the low backscattered power in HCF and provided a spatial resolution of ~ 40 m with a dynamic range (SNR = 1) higher than 24 dB [1]. Such monitoring capabilities are essential as HCF moves into practical field deployments.

On the other hand, the low Rayleigh backscattering in HCFs can be beneficial in the context of same-wavelength bidirectional (BiDi) transmission, which can maximize fiber utilization and simplify network design. In [6-8], high-capacity offline and real-time BiDi transmission experiments using HCFs have been reported. These include our real-time C-band BiDi transmission over

107.5-km field-deployable HCF cable [7], which constitutes the longest BiDi HCF transmission to date. Specifically, we investigated the synergy between low-cost energy-efficient 400G ZR optics and BiDi HCF transmission. Full C-band BiDi transmission ($2 \times 64 \times 400$ Gb/s/ λ) over 107.5-km field-deployable HCF cable was demonstrated under both 27.5-dBm and 34.5-dBm launch powers. Similar performance was achieved in both directions and only minor degradation near the strongest gas-line absorption (GLA) region was observed. Furthermore, the performance of BiDi-HCF transmission was compared with that of unidirectional (UniDi) HCF transmission and a bidirectional back-to-back link (with an equivalent loss emulated by a variable optical attenuator). Negligible penalty was experienced relative to these two cases, confirming the suitability of using HCF for same-wavelength BiDi transmission [7]. Fig. 1(c) summarizes recent demonstrations of ≥ 100 -km straight-line HCF transmission, which highlights our works about real-time HCF transmission relative to the state-of-the-art.

Multi-Span Long-Distance and Long-Haul Transmission

Scaling from single-span metro to long-distance transmission will involve the use of in-line amplifiers (ILAs), as shown in Fig. 1(b). For HCF systems, a distinctive impairment is GLA which in the C-band is primarily induced by absorption from the residual CO₂ in the hollow core. GLA results in wavelength-localized absorption features that lead to impairments which increase with and ultimately limit the transmission distance [1-4].

To eliminate the GLA impact, fiber purging can be employed [25] but may not scale to large scale production since it is time-consuming. Alternatively, reducing the relative data rate at channels affected by GLA [2] and/or varying the channel baud rate in the GLA spectral region to avoid the GLA peaks [2, 19] have also been proven to be effective. However, these will inevitably reduce the overall throughput of the system. In [14], we have proposed and experimentally

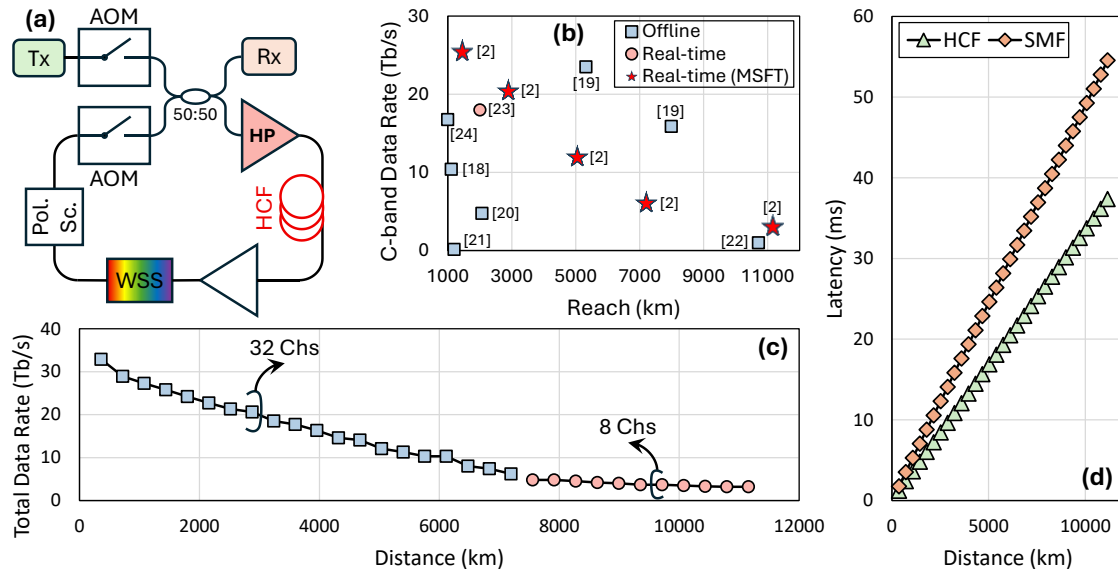


Fig. 1: (a) Illustration of an optical recirculating loop, (b) summary of recent ≥ 1000 km HCF transmission using recirculating loop, (c) total data rate versus distance reported in [2], and (d) latency comparison between HCF and SMF over distance.

demonstrated a simple yet effective transmitter power optimization (TPO) scheme to combat the GLA impact in multi-span long-distance HCF transmission without sacrificing transmission capacity. It was validated in a 3-span 442.66-km HCF system, which transmitted 32×800 Gb/s/λ real-time C-band signals. One HP BA and two HP ILAs provided 34.5-dBm launch power to each span. By optimizing the per-channel transmitter power to create a non-uniform launch profile that favours the GLA-affected channels, uniform performance with >1.5 -dB Q-margins were achieved across all the 32 C-band channels, which represents the longest straight-line fully-loaded C-band HCF transmission to date [14].

To further investigate the performance of HCF transmission over the scale of thousands of kilometres, optical recirculating loop is normally used [2, 18-24]. Fig. 2(a) illustrates a basic loop setup, which uses two acousto-optic modulators (AOMs) to control the circulating light. A pair of complementary pulses is used to control these AOMs so that light can be confined inside the loop to emulate long-haul HCF transmission. Fig. 2(b) shows a summary of recent $>1,000$ km HCF loop transmission. It is seen that the flexibility of offline digital signal processing (DSP) has allowed transmission demonstrations over transoceanic distances. In contrast, our work in [2] explored quasi-real-time loop transmission up to 11,154 km by employing a modified commercial transceiver to synchronise real-time DSP with the loop timing. This allowed investigating long-haul scenarios with similar DSP capability and OSNR requirements as commercial real-time long-haul systems. Fully-loaded C-band transmission up to

7,196 km was explored, before an 8-channel subset of GLA-free channels was used to probe the ultimate distance limits. The total data rate versus transmission distance is depicted in Fig. 2(c). Up to 25.8 Tb/s at 1,439.2 km, 20.6 Tb/s at 2,878.4 km, and 10.3 Tb/s at 6,116.6 km in the 32-channel case were achieved. In the 8-channel configuration, 3.2 Tb/s was demonstrated at 11,154 km. In the meantime, as shown in Fig. 2(d), HCF offers consistent latency reductions ($\sim 31.49\%$) and can bring in >15.5 -ms latency reduction relative to SMF when the reach exceeds 10,000 km.

Discussions and Conclusions

While GLA represents a limiting factor for HCF transmission towards long distances, ongoing advances in HCF design and fabrication are already delivering further significant reductions in both fiber loss and GLA level. In addition, a range of system-level solutions can be explored to mitigate the GLA impact, including flexible channel data rate and/or grid allocation, pre- or post-compensation, and HCF-dedicated advanced DSP. On the other hand, it is also possible to adjust the design of HCFs to achieve spectral window tunability [3], which potentially can open up new transmission bands for future high-speed optical transmission.

In conclusion, recent advances in HCF transmission clearly demonstrate its readiness for metro and regional long-distance deployments and highlight its near-term transformational impact. As these advances continue, HCF has the full potential to underpin next-generation of scalable, high-performance metro, regional and long-haul optical networks.

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References

- [1] A. Ali, Y. Hong, M. Kamalian-Kopae, et al., "Unrepeated HCF Transmission over spans up to 301.7 km," in *Proc. of OFC*, San Francisco, United States, Th4A.3, 2025. DOI: [10.1364/OFC.2025.Th4A.3](https://doi.org/10.1364/OFC.2025.Th4A.3)
- [2] Y. Hong, A. Ali, M. Kamalian-Kopae, et al., "Real-Time, Fully-Loaded C-band, Low-Latency, Long-Haul Transmission over Hollow-Core Fiber," in *Proc. of ECOC*, Copenhagen, Denmark, Th.03.02.2, 2025. DOI: [10.1109/ECOC66593.2025.11262952](https://doi.org/10.1109/ECOC66593.2025.11262952)
- [3] M. Petrovich, E.N. Fokoua, Y. Chen, et al., "Broadband Optical Fibre with An Attenuation Lower than 0.1 Decibel per Kilometre," *Nature Photonics*, vol. 19, pp. 1203–1208, 2025. DOI: [10.1038/s41566-025-01747-5](https://doi.org/10.1038/s41566-025-01747-5)
- [4] Y. Ding, P. Li, G. Chen, et al., "Support Tube Hollow-Core Fiber with 0.05 dB/km Attenuation," in *Proc. of ECOC*, Copenhagen, Denmark, Tu.04.01.2, 2025. DOI: [10.1109/ECOC66593.2025.11263260](https://doi.org/10.1109/ECOC66593.2025.11263260)
- [5] C. Yang, C. Li, Y. Gong, et al., "C+L-band 4 Tb/s (500 Gb/s/λ × 8λ) WDM IM/DD Optical Interconnection over Anti-resonant Hollow-core Fiber Enabled by Ultra-high Bandwidth TFLN Modulator," in *Proc. of ECOC*, Frankfurt, Germany, Th1B.2, 2024. DOI: [10.1109/ECOC55286.2024.10926666](https://doi.org/10.1109/ECOC55286.2024.10926666)
- [6] M. Zuo, D. Wang, D. Ge, et al., "C-Band 2dir.x40λx224 Gb/s Co-Wavelength Bidirectional IM-DD Fronthaul Over 10 km Low-Latency Hollow-Core Fiber," in *Proc. of ECOC*, Copenhagen, Denmark, Th.01.07.3, 2025. DOI: [10.1109/ECOC66593.2025.11263045](https://doi.org/10.1109/ECOC66593.2025.11263045)
- [7] Y. Hong, A. Ali, M. Kamalian-Kopae, et al., "Bidirectional Full C-band Transmission over Hollow-core Cable using 400G ZR," in *Proc. of OFC*, Los Angeles, United States, M1B.4, 2026.
- [8] L. Feng, A. Zhang, J. Luo, et al., "Demonstration of Single-span 100km Hollow Core Fiber Bidirectional Transmission with 1Tb/s/λ Real-time Signals," in *Proc. of OFC*, San Francisco, United States, M1H.6, 2025. DOI: [10.1364/OFC.2025.M1H.6](https://doi.org/10.1364/OFC.2025.M1H.6)
- [9] J. Sun, D. Wang, M. Zuo, et al., "Nonlinearity-Free DP-144QAM-PCS Beyond-1Tbps Transmission over 100km AR-HCF for OTN-based Decentralized Intelligent Training," in *Proc. of OFC*, San Francisco, United States, M1H.5, 2025. DOI: [10.1364/OFC.2025.M1H.5](https://doi.org/10.1364/OFC.2025.M1H.5)
- [10] X. Gao, Q. Zhang, L. Feng, et al., "First Demonstration of Distributed Characterization over 100 km Anti-Resonant Hollow-Core Fiber in Real-time widened C+L-band WDM Transmission," in *Proc. of OFC*, San Francisco, United States, Th3F.7, 2025. [10.1364/OFC.2025.Th3F.7](https://doi.org/10.1364/OFC.2025.Th3F.7)
- [11] T. Dou, S. Gao, Z. Li, et al., "40 Tbps Classical Communication Coexistence with Quantum Key Distribution Over Hundred-Kilometer Hollow-Core Fiber," *Photonics Research*, vol. 14, no. 4, pp. 1658-1669, 2026. DOI: [10.1364/PRJ.582266](https://doi.org/10.1364/PRJ.582266)
- [12] S. Liu, D. Ge, Q. Qiu, et al., "10.7-Tb/s Unrepeated WDM Transmission with >1.2-Tb/s/λ signals over 217.1-km AR-HCF by EDFA-only Amplification," in *Proc. of ACP*, Jiangsu, China, 2025. DOI: [10.1109/ACP66871.2025.11350531](https://doi.org/10.1109/ACP66871.2025.11350531)
- [13] S. Liu, D. Ge, Q. Qiu, et al., "Tb/s/λ-Class Co-Frequency Co-Time Full-Duplex WDM Transmission Over 100-km AR-HCF in the S+C+L-band," *Journal of Lightwave Technology*, vol. 44, no. 6, pp. 2076-2086, 2026. DOI: [10.1109/JLT.2026.3651633](https://doi.org/10.1109/JLT.2026.3651633)
- [14] Y. Hong, M. Kamalian-Kopae, A. Ali, et al., "Transmitter Power Optimization for Uniform Performance Multi-Span Hollow-Core Fiber Transmission," in *Proc. of OFC*, Los Angeles, United States, Th1J.5, 2026.
- [15] L. Deng, S. Chen, Z. Qian, et al., "First Demonstration of 691.2 Gb/s/λ Single-Unrepeated Transmission over 550-km Support-Tube Hollow Core Fiber," in *Proc. of ACP*, Jiangsu, China, 2025. DOI: [10.1109/ACP66871.2025.11350655](https://doi.org/10.1109/ACP66871.2025.11350655)
- [16] L. Feng, A. Zhang, P. Li, et al., "Real-Time 800G/λ Unrepeated Transmission over 397.3km HCF together with 128km SMF," in *Proc. of ACP*, Jiangsu, China, 2025. DOI: [10.1109/ACP66871.2025.11350441](https://doi.org/10.1109/ACP66871.2025.11350441)
- [17] L. Feng, A. Zhang, P. Li, et al., "Real-time Unrepeated Transmission of 400G/800G/1.2T over 726.1km/611.9km/436.1km HCF only Using EDFA," in *Proc. of OFC*, Los Angeles, United States, M2C.3, 2026.
- [18] H. Mardoyan, R.S. B. Ospina, C.C. Carrero, et al., "222-km-long Hybrid Span Transmission Systems made of Support Tube Hollow Core Fiber and of Standard Single Mode Fiber using High Power Doped Fiber Amplifier," in *Proc. of OFC*, Los Angeles, United States, Th1J.4, 2026.
- [19] R. Boddada, C.C. Carrero, H. Mardoyan, et al., "Sparsely repeated 21.7 Tb/s Net-Rate Transoceanic Transmission with 266 km Ultra-Long Spans Enabled by Low IMI and Low loss Hollow Core Fiber," in *Proc. of OFC*, Los Angeles, United States, Th4B.7, 2026.
- [20] A. Nespola, S.R. Sandoghchi, L. Hooper, et al., "Ultra-Long-Haul WDM PM-16QAM Transmission in a Reduced Inter-Modal Interference NANF," in *Proc. of IPC*, Orlando, United States, WB1.3, 2023. DOI: [10.1109/IPC57732.2023.10360760](https://doi.org/10.1109/IPC57732.2023.10360760)
- [21] L. Feng, L. Sun, J. Cai, et al., "Enabling 1200-km Optical DNANF Transmissions via the Space-time Coded Digital Subcarrier Modulation," *Optics Letters*, vol. 50, no. 13, pp. 4170-4173, 2025. DOI: [10.1364/OL.564128](https://doi.org/10.1364/OL.564128)
- [22] D. Ge, S. Liu, Q. Qiu, et al., "1-Tb/s/λ Transmission over Record 10714-km AR-HCF," in *Proc. of ECOC*, Copenhagen, Denmark, W.03.05.5, 2025. DOI: [10.1109/ECOC66593.2025.11263178](https://doi.org/10.1109/ECOC66593.2025.11263178)
- [23] A. Saljoghei, M. Qiu, S.R. Sandoghchi, et al., "First Demonstration of Field-Deployable Low Latency Hollow-core Cable Capable of Supporting >1000km, 400Gb/s WDM Transmission," *arXiv*, 2021. DOI: [10.48550/arXiv.2106.05343](https://doi.org/10.48550/arXiv.2106.05343)
- [24] H. Chen, X. Zhang, S. Jin, et al., "Demonstration of High-Capacity WDM Long-Haul Transmission based on a Long-Span Nested Antiresonant Nodeless Fiber," *Chinese Optics Letters*, vol. 23, no. 9, pp. 090601, 2025. DOI: [10.1364/COL.23.090601](https://doi.org/10.1364/COL.23.090601)
- [25] Y. Xiong, R. Zhao, D. Ge, et al., "CO2 Elimination in Hollow-Core Fibre via Post-Processing," in *Proc. ECOC*, Copenhagen, Denmark, W.02.01.08, 2025. DOI: [10.1109/ECOC66593.2025.11263127](https://doi.org/10.1109/ECOC66593.2025.11263127)